

# BCYC — Biology Deep-Dive (2026-08-24)

## BCYC / Bicycle Therapeutics — Biology Deep-Dive & Burn-Destination Analysis

**2026-08-24 · companion to BCYC\_ADJUDICATION\_20260824.md — the adjudicated record (MINIMUM STARTER 0.25%, gate ≤\$4.30 DAY-only, instr #100) stays authoritative; this memo informs, it does not re-rule.**

Sources: 10-Q filed 2026-07-30 (period 6/30/26, EDGAR 0001104659-26-088457); Q2 press release (EX-99.1, same accession); Q1 8-K/PR (4/30, 0001104659-26-052107); ASCO data PR (5/21, 0001104659-26-065231); director-resignation 8-K (8/20, event 8/18); AGM 8-K (6/18); ClinicalTrials.gov API v2 records NCT06225596, NCT07450859, NCT04561362, NCT04180371, NCT05163041, NCT03486730, NCT07681674 (pulled 2026-08-24); PubMed PMIDs inline. Anything not traceable to these is marked UNVERIFIED.

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### 0. The finding that reframes the principal's question

The question was: *“the cash will be drawn to zero in support of it — I need to understand the biology.”* The filings say the cash is **not** being drawn in support of the asset most people associate with BCYC. In **March 2026** the company executed a strategic reprioritization (10-Q, Note on restructuring): a **30% workforce reduction**, discontinuation of zelenectide's breast and NSCLC trials, **conversion of Duravelo-2 from a Phase 2/3 registrational trial to a randomized Phase 2**, and zelenectide **“deprioritized for internal development while we evaluate next steps for the program following preliminary feedback from regulatory agencies”** (10-Q, verbatim). The cash now funds a different, cheaper bet: **nuzefatide pevedotin (EphA2) and the radioconjugate platform**. The adjudication's decision to carry zelenectide at zero is not just conservatism — it matches management's own capital allocation.

### 1. Mechanism primer — what a Bicycle is and why the format is the whole argument

- A **Bicycle molecule** is a fully synthetic short peptide (~1.5–2 kDa) constrained by a small-molecule scaffold into two loops (company description, 10-Q “About” language; design paper PMID 36112771). An antibody is ~150 kDa — two orders of magnitude larger.

- **Consequences of small size:** rapid, deep tumor penetration; **renal clearance with a short plasma half-life** — the toxin spends little time circulating. An antibody-drug conjugate circulates for ~1–2 weeks per dose, releasing payload systemically the entire time.
- **A Bicycle Toxin Conjugate (BTC/BDC)** = Bicycle binder + cleavable linker + cytotoxin. Zelenectide's payload is **MMAE** (monomethyl auristatin E — the “vedotin” suffix; PMID 36112771 describes BT8009 as Nectin-4 Bicycle–MMAE).
- **Why this matters more than usual:** enfortumab vedotin (Padcev) is **the same target (Nectin-4) and the same payload (MMAE)** on an antibody. Zelenectide vs enfortumab is therefore a rare *controlled experiment in format*: any tox difference is attributable to the delivery scaffold, not the warhead or target. The disclosed deltas (below) are the cleanest in-human evidence yet that the short-exposure thesis does something real.

## 2. Zelenectide pevedotin (BT8009) — the asset our prize table zeroes

**Duravelo-2 (NCT06225596)** — registry verified 8/24: official title still reads “Phase 2/3”; the record now lists **PHASE2 only** (the March conversion, registry updated 5/27/26), **ACTIVE\_NOT\_RECRUITING**, est. enrollment 375, start 1/24/24, **primary completion March 2028 (estimated)**. Cohort 1 = previously untreated mUC (two zelenectide arms + an “Arm 3” — comparator identity truncated in the record pull, **UNVERIFIED** whether chemo or EV+P); Cohort 2 = previously treated (mono + combo arms). Primary outcomes: PFS (Cohort 1), PFS/ORR by BICR (Cohort 2).

**Disclosed efficacy (all company disclosures, primary-source quoted):** | Dataset | Population | n | Result | | — | — | — | — | — | — | Duravelo-2 Cohort 1 dose-opt, 6mg dose (ASCO 5/26; Q2 PR) | Untreated mUC, zele+pembro | 26 | ORR 65% unconfirmed; **BICR-confirmed 58% → 62% (16/26)** with one post-cutoff confirmation; mPFS not mature at wk-27 cutoff (7/23/25 data cut) || Duravelo-1 (NCT04561362) update (ASCO 5/26; Q2 PR) | Untreated **cisplatin-ineligible** mUC, 5mg+pembro, **45% ECOG 2** | 22 | ORR 59% unconf / **50% confirmed (23% CR)**, DCR 82%, **mPFS 13.0 months**, mDOR not mature; no G4/G5 drug-related AEs |

**Disclosed toxicity at the 6mg optimal dose (Q2 PR):** peripheral sensory neuropathy **33%**, skin reactions **17%**, eye disorders **10%**, **zero** drug-related hyperglycemia, **zero** severe skin reactions of any grade. The company's cross-trial claim (5/21 PR): skin reactions **~4-fold lower** and neuropathy **~half** the published SOC rates. Comparator context: EV+P's published profile (EV-302, NEJM 2024, PMID 38446675) includes the class's known skin toxicity — up to SJS/TEN black-box severity (PMIDs 40592676, 35274723), neuropathy, hyperglycemia. **Asymmetric-basis caveat: these are cross-trial comparisons at different maturities in overlapping-but-not-identical populations; the desk has not independently recomputed the published SOC rates.** The Duravelo-1 mPFS of 13.0 months in a 45%-ECOG-2 population against EV-302's published 12.5 months in a fitter population is the single most striking number — and also the most cross-trial-fragile.

**The strategic reality:** despite this data, the program is deprioritized. The stated cause is “preliminary feedback from regulatory agencies” — content undisclosed (**UNVERIFIED beyond the phrase itself**; the plausible reading is that a single-arm/comparable-ORR path to approval was closed, forcing a full randomized program the company won't self-fund against an entrenched SOC). “Further data from the randomized Phase 2 are

expected in **2H 2026**” (company guide). RZLT-check: registry PCD is March 2028 — the 2H26 item is a company-elective interim on a deprioritized program, a *different claim class* than completion (consistent, but slippable without registry violation). A **continued-access protocol (BiCAP, NCT07681674, posted 7/26)** now exists for participants remaining on Bicycle trials — wind-down-compatible infrastructure.

### 3. The Padcev question — what zelenectide’s data is actually worth

Enfortumab vedotin owns Nectin-4: approved, ~blockbuster, and since EV-302 the 1L standard with pembrolizumab. Zelenectide’s honest differentiation claim after the 2026 disclosures is **efficacy in the SOC’s neighborhood with a categorically different tolerability profile** — not superiority. Who pays for that? (a) patients who can’t tolerate EV (the ECOG-2-heavy Duravelo-1 cohort is exactly that population, and 50% confirmed ORR + 13.0mo mPFS there is genuinely clinically meaningful); (b) a partner with urothelial infrastructure willing to fund the randomized program FDA apparently requires; (c) post-EV progression (nuzefatide’s n=14 cohort, 10/14 post-EV, is Bicycle’s own probe of that space — with a *different target*). The market for “similar efficacy, kinder drug” is real but it is a *commercial fight against an entrenched incumbent* — which is why the company chose not to fund it alone, and why the adjudication’s zero is a defensible carrying value with the 2H26 randomized readout as a free call option.

### 4. What the cash IS buying — nuzefatide and the radio platform

- **Nuzefatide pevedotin (formerly BT5528): EphA2-targeting BDC + MMAE.** EphA2 is the target the Q2 PR calls “historically considered undruggable using antibody-based approaches” — an antibody ADC against EphA2 (MedImmune’s MEDI-547) failed on tox years ago; the small-format/fast-clearance design is the mechanistic answer to exactly that failure mode. Evidence stack, all primary-sourced: Ph1/2 (NCT04180371, n=288 enrolled, **PCD 7/31/26 — final dataset now due**); the disclosed mUC post-CPI cohort (6.5mg/m<sup>2</sup> Q2W + nivolumab, **n=14, 10 post-enfortumab**): “differentiated safety profile as well as promising anti-tumor activity” — **no ORR disclosed (UNVERIFIED — the load-bearing efficacy number for the pivot asset is not yet public in a filing**; AACR poster not fetched); PDAC preclinical: EphA2 expressed in **16/16** PDX models, **10/14 sensitive** (6 highly); human gallium-68 EphA2 imaging in 7 PDAC patients (DKTK/DKFZ, AACR) demonstrating target engagement in the indication.
- **The pivot trial: NCT07450859** — Phase 2, nuzefatide **monotherapy** in metastatic PDAC, **n=39**, ORR primary, first patient dosed April 2026 at 8mg/m<sup>2</sup> Q2W, RECRUITING (updated 8/13/26), PCD March 2029. This is a small, cheap, single-arm shot in an indication with essentially no targeted competition and a brutal base rate — the §BIOTECH-BASE-RATE 5–7% the adjudication priced is *this*.
- **Radioconjugates (BRC): BT1702, an MT1-MMP-targeting radioligand — the same target as BT1718**, Bicycle’s discontinued first toxin conjugate (NCT03486730, completed n=72, never advanced). The imaging data validates targeting; the therapeutic bet is that fast-clearing binders are *better* suited to radioligands (deliver dose, clear before marrow soak) than to toxins. Ph1 expected **2027** (company guide; no NCT yet — discretionary spend until started).

- **BT7480 (Nectin-4/CD137 TICA):** NCT05163041 ACTIVE\_NOT\_RECRUITING, PCD August 2026 — parked; a data-package asset, not a spend line.
- **Partnership validation with real cash:** cumulative collaboration payments **\$240.2M** (10-Q): Bayer \$46.8M, Novartis \$53.0M, Ionis \$49.7M, Genentech \$56.0M — all four still recognized in Note 8 as live arrangements. The radio/discovery platform is partially externally funded, which is what makes the “into 2030” runway arithmetic work at a ~\$130M organization.

## 5. Burn decomposition — where the dollars go (the principal’s exact question)

| Item                                     | Number                                                                                                                                                                | Source          |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Cash 12/31/25 → 3/31/26 → 6/30/26        | \$628.1M → \$559.5M → <b>\$510.1M</b>                                                                                                                                 | 10-Q, Q1/Q2 PRs |
| H1-26 operating cash use                 | <b>\$115.5M</b> (vs \$159.2M H1-25, <b>-27%</b> )                                                                                                                     | 10-Q            |
| Q2 R&D / G&A / net loss                  | \$41.2M / \$14.0M / \$50.3M (R&D <b>-\$29.8M</b> y/y)                                                                                                                 | Q2 PR           |
| Largest single driver of the R&D decline | <b>zelenectide</b> clinical spend <b>-\$24.8M</b> (Q2 y/y), <b>-\$36.1M</b> (H1)                                                                                      | 10-Q MD&A       |
| March restructuring                      | ~\$6.5M total severance (+\$3.9M contract terminations, paid)                                                                                                         | 10-Q            |
| Company runway guide                     | <b>“into 2030”</b> (both Q1 and Q2 PRs)                                                                                                                               | PRs             |
| Formal 10-Q statement                    | ≥12 months from filing (standard ASC 205-40 scope)                                                                                                                    | 10-Q            |
| Committed vs discretionary               | Trials running (Duravelo-2 enrolled-and-closed; PDAC n=39; wind-downs) = committed; <b>BT1702 Ph1 (2027) and any zele restart = discretionary, not yet contracted</b> | registry + PRs  |

Trajectory check on the adjudication’s runway band (Q4-2028 at the Q2 rate → mid-2030 at street’s implied decline): Q2’s \$49.4M quarter already includes wind-down costs that fall away (dose-opt fully enrolled, discontinued trials completing, severance mostly paid) — the *direction* supports the long end, but a PDAC expansion or BT1702 start pulls it back. **The pre-mortem’s tripwire-less risk is precisely this: each new trial start converts “runway into 2030” from plan into contract.**

## 6. Failure-mode library — small-format conjugates

- **Melflufen (Oncopeptides)** — peptide-drug conjugate, approved then effectively withdrawn in the US after the OCEAN trial showed an overall-survival disadvantage in a key subgroup (PMIDs 39246181, 36417082). Failure mode: *efficacy-tox balance inverted at commercial dosing; accelerated approval bar not sustained*. Rename-or-retire for zelenectide: partially retired — zelenectide’s randomized Ph2 exists precisely because FDA would not accept a single-arm bar; whatever reads out in 2H26 is against a randomized control, not a house-picked comparator.
- **MEDI-547 (anti-EphA2 antibody ADC)** — dose-limiting bleeding/coagulation tox, terminated Phase 1. Failure mode: *antibody-duration exposure of an EphA2 toxin*. This is the failure nuzefatide’s format is designed to retire, and the n=14 “differentiated safety” claim is the first in-human evidence — but the efficacy leg is undisclosed, so the retirement is half-proven.
- **BT1718 (Bicycle’s own MT1-MMP BTC)** — completed Phase 1/2 (n=72), never advanced; no pivotal data ever presented. Failure mode: *fast clearance can also mean insufficient tumor exposure — the same property that helps tox can starve efficacy*. Zelenectide’s 58–62% confirmed 1L ORR is the counter-evidence for Nectin-4; for EphA2 and for radioligands the question is open. Note the honest circle: BT1718’s target is being re-run as a radioligand (BT1702), where fast clearance is an asset — the platform learned from its own failure mode.
- **Class lesson for sizing:** peptide conjugates have never yet produced an approved oncology drug from this specific format family. The base-rate discipline in the adjudication (§BIOTECH-BASE-RATE, no conditional delta granted) is the correct posture until the PDAC ORR or the randomized zeke data breaks the pattern.

## 7. Differentiated view, sizing relevance, and armed sensors

**Differentiated-view one-liner:** the tape prices a busted-ADC story burning cash to zero; the filings describe a company that already killed its own losing program, cut burn 27%, and now spends ~\$50M/qtr on a first-in-class EphA2 shot (n=39, answer by 2029) plus a radioligand platform with \$240M of external validation — while the “busted” asset still carries randomized 1L data with a genuine format-tox edge that reads out in 2H26 at zero carrying value to us and to management.

**What licenses the next rung (through EXISTING gates only — the EC’s add-gates stand):** - The EC’s three add-gates (capital-return 8-K → recourt; Armistice 13G→13D; Q3 cash ≥\$470M w/ opex ≤\$40M/qtr) are unchanged by this memo. - **Biology-side pre-writable rubric for the 2H26 zelenectide randomized data** (pre-registered now, before any data): PASS = randomized Cohort-1 PFS in the control’s neighborhood with the tox delta intact → the partnering asset is real → court to re-value the zero (not an auto-add). FAIL = control clearly outperforms → the zero was right; no action, since we carry it at nothing. **The readout is upside-skewed by construction** — this is the asymmetry the adjudication priced. - **Nuzefatide efficacy disclosure** (the missing number): any filed ORR from the n=14 mUC cohort or early PDAC signal = the first evidence-class event that could justify a conditional delta above the 5–7% base rate → court.

**Armed sensors (dates primary-verified):** BT5528-100 final data due (registry PCD 7/31/26 — passed; disclosure watch live); BT7480-100 PCD **Aug-2026** (this month — disposition tell); zeke 2H26 randomized interim (company guide, elective, slippable); Duravelo-2 registry PCD **Mar-2028**; PDAC PCD **Mar-2029**; next print ~10/29

(**UNCONFIRMED**, yfinance); Swanton-class governance departures (8/18 resignation “personal reasons” — watch for repeats); BiCAP continued-access protocol activity (wind-down tell).

**Could not verify (radical honesty):** nuzefatide’s response numbers in the n=14 mUC cohort (no ORR in any filing pulled — the pivot asset’s efficacy is currently a characterization, not a number); the content of the regulatory feedback that demoted Duravelo-2; the identity of Duravelo-2 Cohort-1 Arm 3 (comparator); published EV+P tox rates were not independently recomputed (company’s 4x/2x claims are cross-trial); the 10/29 print date; whether a formal zelenectide partnering process exists beyond “evaluate next steps.”